

Biomimetics

Mimicking Nature For Technological Breakthroughs

Being observant enough to capture the clues offered by nature, and pursuing these with multi-disciplinary research can lead to scientific breakthroughs and engineering wonders



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George de Mestral, a Swiss electrical engineer, invented the hook-and-loop fastener (now inseparable with the name of Velcro) by keenly observing how the seeds of burdock got stuck to his coat and his dog's fur every time they brushed against the plant. That was in 1941.

This is neither the first nor the only example of nature-inspired technologies, but definitely a notable one. Since time immemorial, scientists have been inspired by nature to mimic and surpass its capabilities. Be it vision, hearing or energy-saving, there are numerous things that natural systems perform best.

Why not observe and learn from the workings of nature, and use that

knowledge as the basis for man-made processes and products? This thought has indeed continued to fuel many a wonderful invention, and the trend continues till date.

A look at the current generation brings forth several examples of many advanced and commercially-viable technologies that have been inspired by nature: KPIT Cummins' patent-pending night vision system; Stanford University's Neurogrid; the path-breaking sunscreen and bio-plastic projects taken up by Dr Judith Braganca and her students at BITS Pilani's Goa campus; AutoFlex MARAG's special optical thin-film coating; the Ultracane aid for the visually-impaired; the Libelle High-G flight suit; LogiTek's Marble trackball; the BioAvert object-avoidance software program; the solar cell technology being developed by Dr Akhlesh Lakhtakia and his international team; the self-healing solar cells designed by researchers at the Massachusetts Institute of Technology (MIT) and Purdue University; and ant- and bee-colony algorithms used in computing and artificial intelligence (AI).

By chance or by design?

We come across a lot of nature-inspired technologies in the news. Neural networks based on the working of the human brain, semiconductor chip fabrication techniques inspired by the designs on butterfly wings, aeroplane